Marl Jonson

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Education

San Jose State University **Expected Graduation: May 2027**

Bachelor of Science in Computer Science *San Jose, CA*

- **Coursework:** Data Structures & Algorithms, Operating Systems, Database Systems, Computer Networks, Software Engineering
- **Organizations:** Spartan Analytics, Software & Computer Engineering Society, Responsible Computing Club, Product Management Club

Experience

Nevada National Security Sites May 2026 – Present

Software Engineer Intern *North Las Vegas, NV*

- Kickstarted the creation of a reusable Blazor app template, utility package, and internal wiki using C# and ASP.NET Core Blazor, reducing boilerplate code across 35 apps by centralizing functional components shared between isolated projects into a unified codebase
- Started and completed the independent initiative of migrating 30 apps from .NET 8.x to 10 and from NET Framework 4.x to 4.8 by upgrading NuGet packages/dependencies and resolving build issues on Visual Studio, eliminating urgent EOL technical debt for 12 devs
- Refactored legacy AutoMapper object mappings to Mapster across 25 .NET apps, tracked change requests using ServiceNow, and executed weekly prod deployments after hours to an Azure DevOps pipeline, eliminating $1,500/year in projected commercial licensing liabilities
- Contributed to Agile sprint planning, tracked Azure DevOps work items and user stories, and presented progress during weekly sprints

Hack Your Summer Jul. 2026 – Aug. 2026

Software Engineer Solo Participant *Remote*

- Built Databricks pipelines with PySpark to ingest, clean, and organize healthcare datasets into Bronze, Silver, and Gold Delta Lake tables
- Developed reusable transformations to validate incoming data, remove duplicates, and support reliable incremental processing
- Improved query performance by optimizing Delta Lake tables with partitioning, compaction, and efficient Spark execution strategies
- Collaborated with mentors in the Healthcare track to build scalable data workflows supporting healthcare reporting and analytics

San Francisco Bay Area Rapid Transit District Apr. 2025 – Apr. 2026

Product Management Intern *Oakland, CA*

- Developed Transiteers, a modular Scratch video game introducing youth to careers in public transit
- Interviewed 10+ subject matter experts to record various duties and responsibilities of employees
- Conducted user gameplay testing with various age demographics on-site in Oakland and Berkeley, California
- Collected qualitative user feedback and iteratively redesigned and redeveloped video game modules and features

SJSU Responsible Computing Club Jun. 2025 – Mar. 2026

Software Engineering Ambassador *San Jose, CA*

- Developed and deployed a full-stack web application using React, TypeScript, and backend APIs with version control via GitHub
- Built and maintained responsive web designs using Next.js, Tailwind CSS for styling, and Auth0 for secure authentication
- Created Figma wireframes and user flows for UI/UX prototyping; engaged in bi-monthly Agile sprints and weekly stand-ups
- Performed 5 user interviews using A/B testing to evaluate the usefulness of website features for prospective club members

Projects

Zinnia (HPC Cluster Operations Dashboard) | *C#, ASP.NET Core Web API, Entity Framework Core, SQL Server, SignalR, React, Docker*

- Built an ASP.NET Core Web API simulating HPC cluster nodes, queued jobs, running jobs, failures, and utilization metrics
- Implemented live dashboard updates with SignalR, streaming cluster health, queue length, active jobs, and node status
- Modeled job scheduling state transitions with EF Core and SQL Server, including pending, running, completed, and failed jobs
- Containerized the React dashboard and backend with Docker, supporting local deployment of the full cluster operations simulator

Pollinate (Distributed Event Platform) | *Java, Spring Boot, Kafka, PostgreSQL, Redis, Docker, Kubernetes, OpenTelemetry, Prometheus*

- Built 8 Spring Boot microservices simulating medical devices, exchanging HL7 FHIR events through Apache Kafka
- Implemented retry queues, dead letter queues, idempotent consumers, and distributed transactions for reliable event delivery
- Secured REST and gRPC APIs with OAuth2, JWT authentication, RBAC, and audit logging across distributed services
- Deployed to Kubernetes with Helm, GitHub Actions, OpenTelemetry, Prometheus, and Grafana for CI/CD and observability

Technical Skills

Languages: Java, Python, C#, SQL, TypeScript, Kotlin (novice), Go (novice)

Frameworks & Technologies: Spring Boot, ASP.NET Core, Blazor, React, Next.js, FastAPI, PySpark, Apache Spark, Delta Lake, Databricks, Kafka, PostgreSQL, SQL Server, Redis, Docker, Kubernetes

Developer Tools: Microsoft Azure, Azure DevOps, AWS (novice), GitHub Actions, Terraform, Git, OpenTelemetry, Prometheus, Grafana, REST, gRPC, OAuth2, JWT, Postman, Visual Studio

Certifications: Azure Data Fundamentals (DP-900), Azure Fundamentals (AZ-900), Microsoft 365 Fundamentals (MS-900)